3 A program uses a circular queue to store strings. The queue is created as a 1D array, `QueueArray`, with 10 string items.

The following data is stored about the queue:

- the head pointer initialised to 0
- the tail pointer initialised to 0
- the number of items in the queue initialised to 0.

(a) Declare the array, head pointer, tail pointer and number of items.

If you are writing in Python, include attribute declarations using comments.

Save your program as **Question3_J2022**.

Copy and paste the program code into **part 3(a)** in the evidence document.

[2]

(b) The function `Enqueue` is written in pseudocode. The function adds `DataToAdd` to the queue. It returns `FALSE` if the queue is full and returns `TRUE` if the item is added.

The function is incomplete, there are **five** incomplete statements.

```
FUNCTION Enqueue(BYREF QueueArray[] : STRING, BYREF HeadPointer : INTEGER,
                 BYREF TailPointer : INTEGER, NumberItems : INTEGER,
                 DataToAdd : STRING) RETURNS BOOLEAN

  IF NumberItems = ........................................ THEN

    RETURN ........................................

  ENDIF

  QueueArray[........................................] ← DataToAdd

  IF TailPointer >= 9 THEN

    TailPointer ← ........................................

  ELSE

    TailPointer ← TailPointer + 1

  ENDIF

  NumberItems ← NumberItems ........................................

  RETURN TRUE

ENDFUNCTION
```

Write program code for the function `Enqueue()`.

Save your program. Copy and paste the program code into **part 3(b)** in the evidence document.

[7]

(c) The function `Dequeue()` returns `"FALSE"` if the queue is empty, or it returns the next data item in the queue.

Write program code for the function `Dequeue()`.

Save your program. Copy and paste the program code into **part 3(c)** in the evidence document.

[6]

(d) **(i)** Amend the main program to:

- take as input 11 string values from the user
- use the `Enqueue()` function to add each element to the queue
- output an appropriate message to state whether each addition was successful, or not
- call `Dequeue()` function twice and output the return value each time.

Save your program. Copy and paste the program code into **part 3(d)(i)** in the evidence document.

[5]

(ii) Test your program with the input data:

`"A" "B" "C" "D" "E" "F" "G" "H" "I" "J" "K"`

Take a screenshot to show the output.

Copy and paste the screenshot into **part 3(d)(ii)** in the evidence document.

[1]